

CSE 291G: Physics Simulation Midterm

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1 Acknowledgment

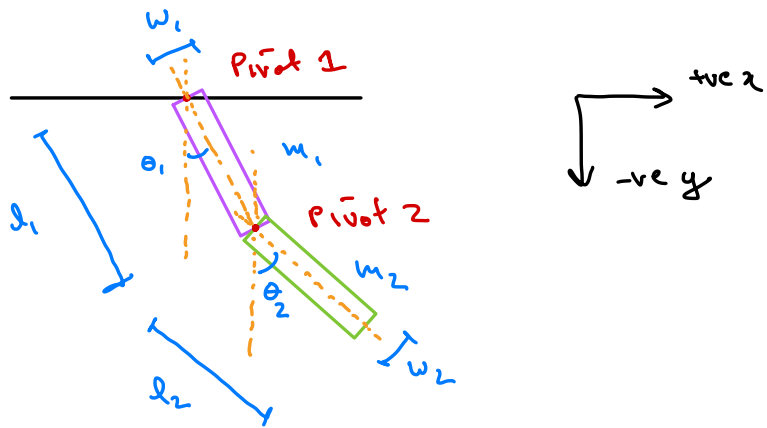
LLMs were used to help with integration and derivation for certain parts of the problem derivation presented below.

2 Project Description and Derivation

For the midterm project, I implemented a double pendulum. Both pendulums are continuous, solid, rigid bodies with an area in the 2D planar space. This requires consideration along the length and width of the pendulum. I have derived the analytical solution using the Least Action Principle (LAP) for this system. The derivation is provided below. RK4 is used as the numerical integrator to generate the simulation using this analytical solution. The system is specifically set to work within the SI units, instead of being dimensionless.

The first pendulum connected to the origin has a length of five meters, a width of one meter, and a mass of one kilogram. The second pendulum has a pivot point at the other end of the first pendulum. The second pendulum is initialized with a length of 3 m, a width of 1.5 m, and a mass of 2 kg. The simulation is run for 30 seconds with the Δt of 0.001 seconds or 1 millisecond.

Starting from the next page, the entire derivation of the double pendulum system is presented. This includes a visualization, the state configuration, the variables of concern, kinetic energy derivation, potential energy derivation, and Euler-Lagrange derivation, which ultimately results in a second-order linear equation. This is reduced to a first-order equation to be used with RK4 to march the simulation.



Configuration: $(\theta_1, \theta_2)^T$

Rod 1 has length l_1 , width w_1 , mass m_1 .
 Rod 2 has length l_2 , width w_2 , mass m_2

Both length & width are measured from pivot points

For rod i , let α_i be the distance of the particle from pivot measured along rod centerline and β_i be the distance measured from the centerline

$$\alpha_i \in [0, l_i]$$

$$\beta_i \in [-w_i/2, w_i/2]$$

Since the mass is uniform, a small particle's mass on a rod is:

$$\sigma_i = \frac{M_i}{l_i w_i} \quad dm_i = \sigma_i d\alpha_i d\beta_i$$

$$q_1 = \begin{pmatrix} \alpha_1 \sin \theta_1 + \beta_1 \sin(90 - \theta_1) \\ -\alpha_1 \cos \theta_1 + \beta_1 \cos(90 - \theta_1) \end{pmatrix}$$

$$= \begin{pmatrix} \alpha_1 \sin \theta_1 + \beta_1 \cos \theta_1 \\ -\alpha_1 \cos \theta_1 + \beta_1 \sin \theta_1 \end{pmatrix}$$

$$q_2 = \begin{pmatrix} l_1 \sin \theta_1 + \alpha_2 \sin \theta_2 + \beta_2 \sin(90 - \theta_2) \\ -l_1 \cos \theta_1 - \alpha_2 \cos \theta_2 + \beta_2 \cos(90 - \theta_2) \end{pmatrix}$$

$$= \begin{pmatrix} l_1 \sin \theta_1 + \alpha_2 \sin \theta_2 + \beta_2 \cos \theta_2 \\ -l_1 \cos \theta_1 - \alpha_2 \cos \theta_2 + \beta_2 \sin \theta_2 \end{pmatrix}$$

$$\dot{q}_1 = \begin{pmatrix} \alpha_1 \sin \dot{\theta}_1 \dot{\theta}_1 + \beta_1 \cos \dot{\theta}_1 \dot{\theta}_1 \\ -\alpha_1 \cos \dot{\theta}_1 \dot{\theta}_1 + \beta_1 \sin \dot{\theta}_1 \dot{\theta}_1 \end{pmatrix}$$

$$\dot{q}_2 = \begin{pmatrix} l_1 \sin \dot{\theta}_1 \dot{\theta}_1 + \alpha_2 \sin \dot{\theta}_2 \dot{\theta}_2 + \beta_2 \cos \dot{\theta}_2 \dot{\theta}_2 \\ -l_1 \cos \dot{\theta}_1 \dot{\theta}_1 - \alpha_2 \cos \dot{\theta}_2 \dot{\theta}_2 + \beta_2 \sin \dot{\theta}_2 \dot{\theta}_2 \end{pmatrix}$$

$$|\dot{q}_1|^2 = \alpha_1^2 \dot{\theta}_1^2 + \beta_1^2 \dot{\theta}_1^2$$

$$|\dot{q}_2|^2 = l_1^2 \dot{\theta}_1^2 + \dot{\theta}_2^2 (\alpha_2^2 + \beta_2^2) + 2 l_1 \dot{\theta}_1 \dot{\theta}_2 [\alpha_2 \cos(\theta_1 - \theta_2) + \beta_2 \sin(\theta_1 - \theta_2)]$$

Kinetic Energy:

$$K_{\text{Total}} = \int \frac{1}{2} |\dot{\mathbf{q}}_1|^2 dm_1 + \int \frac{1}{2} |\dot{\mathbf{q}}_2|^2 dm_2$$

Rod 1:

$$K_1 = \int_0^L \int_{-w/2}^{w/2} \frac{1}{2} (\alpha_1^2 + \beta_1^2) \dot{\theta}_1^2 \sigma_1 d\beta_1 d\alpha_1$$

$$K_1 = \frac{1}{2} \frac{m_1}{L_1 w_1} \dot{\theta}_1^2 \int_0^L \int_{-w/2}^{w/2} (\alpha_1^2 + \beta_1^2) d\beta_1 d\alpha_1$$

The integral can be split into two

$$K_1 = \frac{1}{2} \frac{m_1}{L_1 w_1} \dot{\theta}_1^2 \left(w_1 \cdot \frac{L_1^3}{3} + L_1 \frac{w_1^3}{12} \right)$$

$$= \frac{1}{2} \dot{\theta}_1^2 \left(\frac{m_1 L_1^2}{3} + \frac{m_1 w_1^2}{12} \right) = \frac{1}{2} \dot{\theta}_1^2 I_1$$

$$\text{with } I_1 = \frac{m_1 L_1^2}{3} + \frac{m_1 w_1^2}{12}$$

Rod 2:

$$K_2 = \int_0^L \int_{-w/2}^{w/2} \frac{1}{2} \left(\underbrace{L_1^2 \dot{\theta}_1^2}_{\text{Part A}} + \underbrace{\dot{\theta}_2^2 (\alpha_2^2 + \beta_2^2)}_{\text{Part B}} + \underbrace{2 L_1 \dot{\theta}_1 \dot{\theta}_2 [\alpha_2 \cos(\theta_1 - \theta_2) + \beta_2 \sin(\theta_1 - \theta_2)]}_{\text{Part C}} \right) \sigma_2 d\beta_2 d\alpha_2$$

For simplicity, I am adding the integrals of all these parts directly here. They are trivial to derive

$$\begin{aligned}
 K_2 &= \frac{1}{2} \frac{m_2}{l_2 \omega_2} \left(\overbrace{l_1^2 \dot{\theta}_1^2 l_2 \omega_2^2}^{\text{Part A}} + \overbrace{\frac{m_2 l_2^3 \dot{\theta}_2^2}{3}}^{\text{Part B}} + \overbrace{\frac{l_2 \omega_2^3 \dot{\theta}_2^2}{12} + l_1 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1 - \theta_2) \omega_2 l_2^2)}^{\text{Part C}} \right) \\
 &= \frac{1}{2} \left(m_2 l_1^2 \dot{\theta}_1^2 + \left(\frac{m_2 l_2^2}{3} + \frac{m_2 \omega_2^2}{12} \right) \dot{\theta}_2^2 + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1 - \theta_2)) \right) \\
 &= \frac{1}{2} \left(m_2 l_1^2 \dot{\theta}_1^2 + I_2 \dot{\theta}_2^2 + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1 - \theta_2)) \right) \\
 &\quad \text{with } I_2 = \frac{m_2 l_2^2}{3} + \frac{m_2 \omega_2^2}{12}
 \end{aligned}$$

$$K_{\text{Total}} = K_1 + K_2$$

$$= \frac{1}{2} \dot{\theta}_1^2 I_1 + \frac{1}{2} m_2 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} \dot{\theta}_2^2 I_2 + \frac{1}{2} m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1 - \theta_2))$$

$$K_{\text{Total}} = \frac{1}{2} (I_1 + m_2 l_1^2) \dot{\theta}_1^2 + \frac{1}{2} I_2 \dot{\theta}_2^2 + \frac{1}{2} m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1 - \theta_2))$$

Potential Energy:

$$\int g \, d_{\text{origin}} \, dm_1 + \int g \, d_{\text{origin}} \, dm_2$$

Rod 1:

$$\int_0^{l/2} \int_{-w/2}^{w/2} g (-\alpha_1 \cos \theta_1 + \beta_1 \sin \theta_1) \sigma_1 \, d\beta \, d\alpha$$

$$= \int_0^{l/2} \int_{-w/2}^{w/2} g -\alpha_1 \cos \theta_1 \, \sigma_1 \, d\beta \, d\alpha + \int_0^{l/2} \int_{-w/2}^{w/2} g \beta_1 \sin \theta_1 \, \sigma_1 \, d\beta \, d\alpha$$

→ This evaluates to zero

$$= g \cos \theta_1 r_1 \int_0^l \int_{-w/2}^{w/2} -\alpha \, d\beta \, d\alpha$$

$$= -g \cos \theta_1 \frac{m_1}{l_1 w_1} \cdot w_1 \frac{l_1^2}{2} = -\frac{m_1 g l_1}{2} \cos \theta_1$$

Rod 2:

$$\int_0^l \int_{-w/2}^{w/2} g (-l_1 \cos \theta_1 - \underbrace{\alpha_2 \cos \theta_2 + \beta_2 \sin \theta_2}_{\text{similar to rod 1}}) r_2 \, d\beta \, d\alpha$$

$$= \int_0^l \int_{-w/2}^{w/2} g (-l_1 \cos \theta_1) r_2 \, d\beta \, d\alpha + \left(-\frac{m_2 g l_2}{2} \cos \theta_2 \right)$$

$$= -g l_1 \cos \theta_1 \frac{m_2}{l_2 w_2} l_2 w_2 + \left(-\frac{m_2 g l_2}{2} \cos \theta_2 \right)$$

$$= -m_2 g l_1 \cos \theta_1 + \left(-\frac{m_2 g l_2}{2} \cos \theta_2 \right)$$

$$U_{\text{Total}}(q) = -\left(\frac{m_1}{2} + m_2 \right) g l_1 \cos \theta_1 - \frac{m_2 g l_2}{2} \cos \theta_2$$

or

$$U_{\text{Total}}(q) = -c_1 \cos \theta_1 - c_2 \cos \theta_2$$

$$c_1 = \left(\frac{m_1}{2} + m_2 \right) g l_1 \quad \text{and} \quad c_2 = \frac{m_2 g l_2}{2}$$

Lagrangian: $L(q, \dot{q}) = K - U$

$$\left(\frac{1}{2}(I_1 + m_2 l_1^2) \dot{\theta}_1^2 + \frac{1}{2} I_2 \dot{\theta}_2^2 + \frac{1}{2} m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) - (-c_1 \cos \theta_1 - c_2 \cos \theta_2)$$

Let $a_1 = I_1 + m_2 l_1^2$, $a_2 = I_2$ & $b = \frac{1}{2} m_2 l_1 l_2$

$$\Rightarrow \frac{1}{2} a_1 \dot{\theta}_1^2 + \frac{1}{2} a_2 \dot{\theta}_2^2 + b \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + c_1 \cos \theta_1 - c_2 \cos \theta_2$$

From Least Action Principle

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = 0$$

For θ_1 : $\frac{\partial L}{\partial \theta_1} = -b \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - c_1 \sin \theta_1$

$$\frac{\partial L}{\partial \dot{\theta}_1} = a_1 \dot{\theta}_1 + b \cos(\theta_1 - \theta_2) \dot{\theta}_2$$

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} &= a_1 \ddot{\theta}_1 + [b \cos(\theta_1 - \theta_2) \ddot{\theta}_2 - b \sin(\theta_1 - \theta_2) \dot{\theta}_2 (\dot{\theta}_1 - \dot{\theta}_2)] \\ &= a_1 \ddot{\theta}_1 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_2 - b \sin(\theta_1 - \theta_2) \dot{\theta}_2 \dot{\theta}_1 + b \sin(\theta_1 - \theta_2) \dot{\theta}_2^2 \end{aligned}$$

$$\frac{\partial L}{\partial \theta_1} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} = 0$$

$$(-b \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - c_1 \sin \theta_1 - (a_1 \ddot{\theta}_1 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_2 - b \sin(\theta_1 - \theta_2) \dot{\theta}_2 \dot{\theta}_1 + b \sin(\theta_1 - \theta_2) \dot{\theta}_2^2) = 0$$

$$a_1 \ddot{\theta}_1 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_2 + b \sin(\theta_1 - \theta_2) \dot{\theta}_2^2 + c_1 \sin \theta_1 = 0$$

$$F_{\theta_2}: \frac{\partial L}{\partial \theta_2} = b \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - c_2 \sin \theta_2$$

$$\frac{\partial L}{\partial \dot{\theta}_2} = a_2 \dot{\theta}_2 + b \cos(\theta_1 - \theta_2) \dot{\theta}_1$$

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_2} &= a_2 \ddot{\theta}_2 + [b \cos(\theta_1 - \theta_2) \ddot{\theta}_1 - b \sin(\theta_1 - \theta_2) \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2)] \\ &= a_2 \ddot{\theta}_2 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_1 + b \sin(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2 - b \sin(\theta_1 - \theta_2) \dot{\theta}_1^2 \end{aligned}$$

$$\frac{\partial L}{\partial \theta_1} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} = 0$$

$$b \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - c_2 \sin \theta_2 -$$

$$(a_2 \ddot{\theta}_2 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_1 + b \sin(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2 - b \sin(\theta_1 - \theta_2) \dot{\theta}_1^2) = 0$$

$$a_2 \ddot{\theta}_2 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_1 - b \sin(\theta_1 - \theta_2) \dot{\theta}_1^2 + c_2 \sin \theta_2 = 0$$

Combining Euler Lagrange for theta 1 & 2

$$a_1 \ddot{\theta}_1 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_2 + b \sin(\theta_1 - \theta_2) \dot{\theta}_2^2 - c_1 \sin \theta_1 + a_2 \ddot{\theta}_2 + b \cos(\theta_1 - \theta_2) \ddot{\theta}_1 - b \sin(\theta_1 - \theta_2) \dot{\theta}_1^2 + c_2 \sin \theta_2 = 0$$

This can be written as matrix equation

$$\begin{bmatrix} a_1 & b \cos(\theta_1 - \theta_2) \\ b \cos(\theta_1 - \theta_2) & a_2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \begin{bmatrix} b \sin(\theta_1 - \theta_2) \dot{\theta}_2^2 \\ -b \sin(\theta_1 - \theta_2) \dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} c_1 \sin \theta_1 \\ c_2 \sin \theta_2 \end{bmatrix} = 0$$

$$\text{where } a_1 = I_1 + m_2 l_1^2, \quad a_2 = I_2, \quad b = \frac{1}{2} m_2 l_1 l_2$$

$$c_1 = \left(\frac{m_1}{2} + m_2 \right) g l_1, \quad c_2 = \frac{m_2 g l_2}{2}$$

$$I_1 = \frac{m_1 l_1^2}{3} + \frac{m_1 w_1^2}{12} \quad \text{and} \quad I_2 = \frac{m_2 l_2^2}{3} + \frac{m_2 w_2^2}{12}$$

This takes the standard form:

$$M(q) \ddot{q} + c(q, \dot{q}) + g(q) = 0$$

To solve this using RK4, this needs to be converted to first order

$$\text{Assume } w = \dot{q}$$

$$M(q) \dot{q} = -c(q, \dot{q}) - g(q)$$

$$\dot{q} = -M^{-1}(q) [c(q, \dot{q}) + g(q)]$$

$$\dot{w} = -M^{-1}(q) [c(q, w) + g(q)]$$

$$y(k) = \begin{pmatrix} q(k) \\ \dot{q}(k) \end{pmatrix} \rightarrow \text{state}$$

$\rightarrow$ state-derivative

$$\dot{y}(t) = \begin{pmatrix} \dot{q}(t) \\ \ddot{q}(t) \end{pmatrix} = \begin{pmatrix} \dot{q}(t) \\ M^{-1}(q) [-c(q, w) - g(q)] \end{pmatrix}$$

RK4 is then used to apply the state derivative to state to generate next state